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Updating: Perspectives for HET physics studies at a muon-hadron collider at CERN

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Outline

1. LHμC@CERN: physics motivation and benchmarks
2. HET overview: Higgs, electroweak and top
3. Precision Higgs physics at LHμC
4. High-energy $\gamma\gamma$ physics at LHμC
5. Top physics at LHμC
6. Beyond SM and long-term outlook

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Why a muon–hadron collider for HET physics?

- A muon-hadron collider combines the clean lepton-hadron initial state of DIS with a multi-TeV centre-of-mass energy, opening a new regime for Higgs, electroweak, top, and BSM studies.
- Relative to an electron-proton collider, replacing the electron beam by a muon beam strongly relaxes the synchrotron-radiation limitation, allowing significantly higher beam energies and hence much larger hard-scattering rates, and pushes the centre-of-mass energy deep into the multi-TeV regime.
- Relative to proton-proton collisions, the μp environment remains experimentally cleaner, with smaller QCD backgrounds for many precision and BSM channels.

Three complementary physics paths at LHμC

LHμC combines three powerful and complementary directions: a strong QCD programme, high-precision Higgs/top/electroweak measurements, and a powerful probe of physics beyond the Standard Model.

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Benchmark working points used in this talk

- The present talk focuses on the physics case of the LHμC, using the current HELP4CERN working benchmarks:
- The nominal baseline is

$$E_\mu = 500\ \mathrm{GeV},\ E_p = 7\ \mathrm{TeV},\ \sqrt{s} \simeq 3.7\ \mathrm{TeV},$$

with a $\mathcal{L}_{\mathrm{int}} = 250\ \mathrm{fb}^{-1}$.

- The main upgrade scenario is

$$E_\mu = 1\ \mathrm{TeV},\ E_p = 7\ \mathrm{TeV},\ \sqrt{s} \simeq 5.3\ \mathrm{TeV},$$

with a $\mathcal{L}_{\mathrm{int}} = 1000\ \mathrm{fb}^{-1}$.

LHμC physics message

These working points are sufficient to demonstrate the main message of LHμC: already at the 500 GeV baseline, LHμC enters the multi-TeV regime and becomes highly competitive for Higgs, top, electroweak, $\gamma\gamma$, and representative BSM studies.

Benchmarks

Machine	**Benchmark**
LHeC	$E_e = 20,\ 60$ GeV
LHμC baseline	$E_\mu = 500$ GeV
LHμC upgrade	$E_\mu = 1$ TeV
Configuration	**Value**
$\sqrt{s}_{\mu p}$ (baseline)	$\simeq 3.7$ TeV
$\sqrt{s}_{\mu p}$ (upgrade)	$\simeq 5.3$ TeV
$\mathcal{L}_{\mathrm{int}}$ (baseline)	$250\ \mathrm{fb}^{-1}$
$\mathcal{L}_{\mathrm{int}}$ (upgrade)	$1000\ \mathrm{fb}^{-1}$

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See talk by Krzysztof Piotrzkowski

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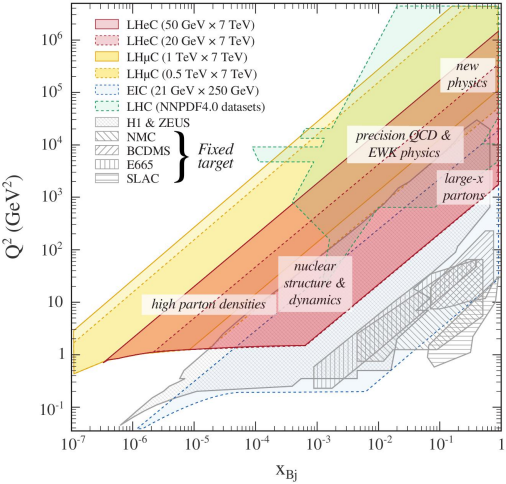
Kinematic reach and QCD physics landscape



Available (x, Q^2) plane for the proposed LHμC, compared with fixed-target, collider, and EIC kinematic coverage.

QCD program at LHμC

- The multi-TeV μp configuration extends the DIS kinematic coverage well beyond HERA and significantly beyond the LHeC.
- The same machine that reaches extreme x and Q^2 also strengthens the QCD programme through improved access to proton PDFs, and precise determinations of α_s.
- This motivates the broader CERN bridge-project narrative: LHμC strengthens both the QCD frontier and the HET/BSM programme.

See talk by Piotr Kotko

Representative HET benchmark cross sections

What does this table show?

LHμC as a genuine HET factory: Higgs production is large enough to support precision Higgs physics, including benchmark coupling studies & the top sector benefits the most, enabling both precision top measurements and sensitivity to rare top processes, & even the 500 GeV baseline opens a qualitatively new regime for electroweak, EFT, and BSM physics.

Benchmark inclusive cross sections

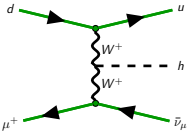
Process	20 GeV LHeC	50 GeV LHeC	500 GeV LHμC	1 TeV LHμC
$\sigma(\ell p \to \nu_\ell H + \text{jet} + X)$ [fb]	52	164	438	731
$\sigma(\ell p \to \ell H + \text{jet} + X)$ [fb]	5.6	19	135	221
$\sigma(\ell p \to \ell\, t\bar{t} + X)$ [pb]	0.0015	0.0323	1.03	2.23
$\sigma(\ell p \to \nu_\ell t + X)$ [pb]	0.55	2.89	21.26	36.77

Key comparisons

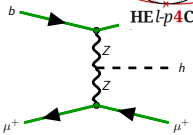
Relative to the 50 GeV LHeC benchmark, the 500 GeV LHμC increases rates by approximately: ~ 2.7 for charged-current Higgs production, ~ 7 for single-top production, and ~ 32 for $t\bar{t}$ production.

Higgs production at LHμC: channels, rates, and statistics

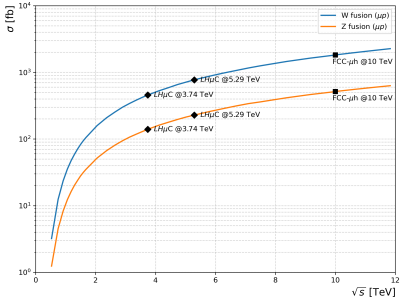
- Higgs production is dominated by CC VBF, with a subleading but still sizeable NC contribution.
- Using the current benchmarks:
  - 500 GeV LHμC: $\sigma_{\rm CC} \simeq 438$ fb, $\sigma_{\rm NC} \simeq 135$ fb,
  - 1 TeV LHμC: $\sigma_{\rm CC} \simeq 731$ fb, $\sigma_{\rm NC} \simeq 221$ fb.
- For the benchmark integrated luminosities, these correspond to roughly:
  - 1.10×10^5 CC Higgs and 3.38×10^4 NC Higgs events at 500 GeV,
  - 7.31×10^5 CC Higgs and 2.21×10^5 NC Higgs events at 1 TeV.
- This already motivates precision Higgs-coupling studies beyond a mere rescaling of LHeC expectations.



CC: $\mu^+ p \to \bar{\nu}_\mu\, j\, h$



NC: $\mu^+ p \to \mu^+\, j\, h$



Higgs boson production cross sections at LHμC for the CC (W-fusion) and subleading NC (Z-fusion) contributions.

Precision Higgs case study: $h \to b\bar{b}$ and coupling prospects

- A concrete benchmark study already exists for the charged-current VBF channel

$$p\mu^+ \to j\,\bar{\nu}_\mu\,h, \qquad h \to b\bar{b}.$$

- Analysis chain used in this study:
  - MG5_aMC@NLO + Pythia6 + FastJet + Delphes,
  - LHeC-like asymmetric detector setup and 75% b-tagging efficiency,
  - selection with exactly two b-jets and

$$|m_{bb} - m_h| < 25\ \mathrm{GeV}.$$

- This gives a benchmark on the measurable rate which can be translated into a sensitivity on g_{hbb}:

$$\sigma(p\mu^+ \to j\,\bar{\nu}_\mu\,b\bar{b}) \;\propto\; \frac{g_{hWW}^2\,g_{hbb}^2}{\Gamma_h},$$

	LHμC-500	LHμC-[illegible]
σ_{sig} [pb]	0.24	0.39
$\sigma_{\mathrm{bgd\text{-}Z}}$ [pb]	0.30	0.50
$\sigma_{\mathrm{bgd\text{-}mj}}$ [pb]	357.9	563.8
$\Delta\sigma/\sigma$	1.19%	0.69%
$\Delta g_{hbb}/g_{hbb}$	0.77%	0.60%

Benchmark charged-current $h \to b\bar{b}$ sensitivities in our study. The quoted coupling reach follows from the benchmark rate measurement after propagating the external uncertainty on g_{hWW}^2/Γ_h.

Key finding

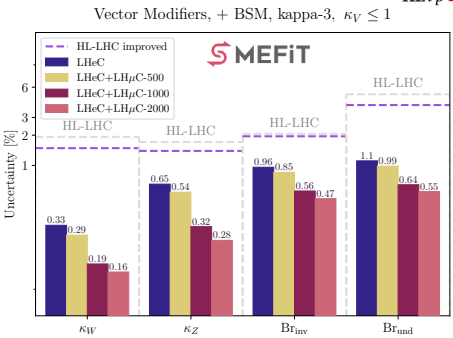
The current LHμCbenchmark study already reaches sub-percent sensitivity to g_{hbb} in the charged-current Higgs channel, turning the large production rates into a concrete precision-Higgs case.

Broader Higgs-coupling reach: a global-fit picture

- Beyond the dedicated $h \to b\bar{b}$ benchmark, we also examined a Higgs coupling fit in the kappa-3 framework.
- These results are obtained in a first explorative study:
  - the same systematic factors as in the underlying LHeC-based setup,
  - the statistical error is rescaled using the benchmark LHμC cross sections,
  - and the fit results shown here were obtained with the SMEFiT framework.
- The strongest visible improvement is in the vector sector:

$$\delta\kappa_W \simeq 0.29\% \ (\mathrm{LH}\mu\mathrm{C\text{-}0.5TeV}),\ 0.19\% \ (\mathrm{LH}\mu\mathrm{C\text{-}1TeV}).$$

- The clearest message of the current exploratory fit is the strong projected improvement in the vector sector, especially for κ_W and κ_Z, at the higher-energy LHμCstages.



Exploratory vector-modifier result in the kappa-3 framework. The panel compares HL-LHC-based benchmarks with LHeC and with LHeC+LHμC stages.

Electroweak programme and EFT opportunities

EW precision reach at LHμC

- Beyond Higgs rates, LHμC offers a broader electroweak programme based on high-Q^2 DIS, vector-boson fusion, and photon-induced reactions.
- The larger kinematic span relative to the LHeC is expected to strengthen sensitivity to key electroweak observables, in particular m_W and $\sin^2\theta_W$, while also extending the SMEFT lever arm.
- Overall, LHμC is expected to extend the precision electroweak reach beyond that projected for the LHeC.

Highlighted EW / EFT opportunities

- High-scale measurements of the weak mixing angle and its running,
- Precision light-quark weak neutral-current couplings via γZ interference and Z exchange,
- Improved sensitivity to m_W in a clean DIS environment,
- Electroweak/top measurements linked to the Wtb vertex and direct sensitivity to V_{ts},
- SMEFT directions complementary to those probed at pp and e^+e^- machines,
- Photon-induced EFT sensitivity in $\gamma\gamma \to W^+W^-$, ZZ, and heavy final states.

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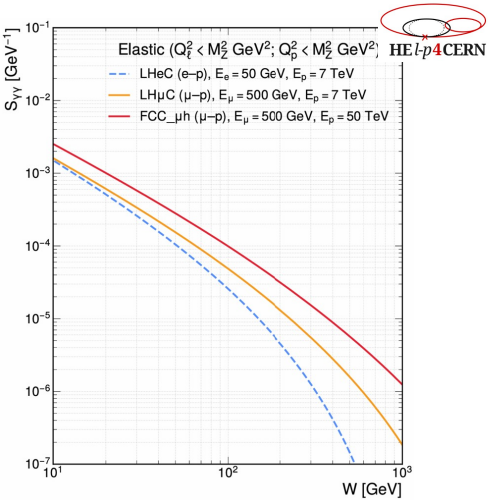
High-energy $\gamma\gamma$ physics is a natural part of the LHμC programme

Interesting HET directions

- LHμC naturally operates as a high-energy $\gamma\gamma$ collider.
- In the 500 GeV baseline, the effective $\gamma\gamma$ centre-of-mass energy extends into the multi-TeV regime, opening a broad programme of clean exclusive measurements.
  - anomalous τ moments,
  - $\gamma\gamma \to W^+W^-$ and anomalous quartic gauge couplings (qQGC),
  - loop-induced $\gamma\gamma \to ZZ$,
  - exclusive $t\bar{t}$ production,
  - light-by-light scattering (LbL)
  - BSM charged-particle pair production.

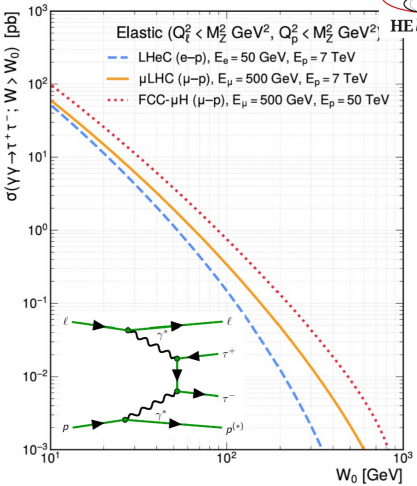


Elastic photon-photon luminosity spectrum $S_{\gamma\gamma}$ as a function of the $\gamma\gamma$ invariant mass W, comparing the LHeC, LHμC, and FCC-μh benchmark configurations.

Flagship $\gamma\gamma$ channel: $\tau^+\tau^-$ and anomalous τ moments

LHμC $\tau^+\tau^-$ physics message

- The elastic $\gamma\gamma \to \tau^+\tau^-$ rate at LHμC for $W_0 = 10$ GeV is already very large, with $\sigma_{\rm el} \simeq 58.5$ pb at 3.7 TeV and about 70.3 pb at the 5.3 TeV upgrade benchmark.
- This provides a high-statistics probe of the $\gamma\tau\tau$ vertex and anomalous magnetic (a_τ) and electric (d_τ) dipole moments.
- Anomalous contributions harden the high-$M_{\tau\tau}$ tail, making the invariant-mass spectrum the key discriminating observable.
- This is one of the cleanest examples of how LHμC combines high energy with a low-background exclusive topology.



Elastic $\gamma\gamma \to \tau^+\tau^-$ cross section above the threshold W_0, comparing the LHeC, LHμC, and FCC-μh benchmark configurations.

Exclusive $\gamma\gamma$ channels: electroweak, top, and BSM reach

$\gamma\gamma \to W^+W^-$

Strong aQGC sensitivity comes from the high-M_{WW} tail

$\gamma\gamma \to ZZ$

Loop-induced in the SM; clean probe of high-mass electroweak dynamics

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Exclusive high-energy $\gamma\gamma$ programme

$\gamma\gamma \to t\bar{t}$

Probes heavy thresholds and top-sector dynamics in exclusive topology

Exclusive BSM pairs

Charged states such as higgsinos and sleptons

Beyond $\tau^+\tau^-$, the high-energy $\gamma\gamma$ programme at LHμC spans electroweak, top, and BSM-sensitive channels in a clean exclusive environment.

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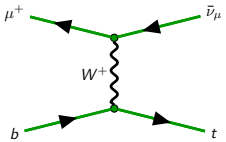
Top physics at LHμC: production mechanisms and experimental promise

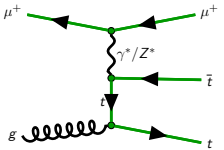
- The top sector is one of the clearest HET advantages of LHμC relative to the LHeC.
- **Single-top production** is dominated by charged-current DIS, with representative subprocess $\mu^+ b \to \bar{\nu}_\mu t$, mediated by t-channel W^+ exchange in the 5-flavor scheme.
- **Top-pair production** is dominated by neutral-current lepton–gluon scattering, with representative process $\mu^+ g \to \mu^+ t\bar{t}$, driven mainly by $\gamma^* g \to t\bar{t}$, while $Z^* g \to t\bar{t}$ is subleading.
- The top advantage is not only a formal cross-section enhancement: it is also supported by favourable acceptance, making the programme experimentally promising after realistic selections.

LO CC single-top production



CC DIS with t-channel W^+ exchange (5F scheme) with inclusive rates: $\sigma = 21.26$ pb, and 36.77 pb.

LO NC top-pair production

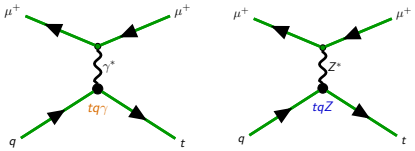


Dominant hard subprocess: $\gamma^* g \to t\bar{t}$; $Z^* g \to t\bar{t}$ is subleading with inclusive rates: $\sigma = 1.03$ pb, and 2.23 pb.

Top-FCNC probes at LHμC: $tq\gamma$ and tqZ

- In the Standard Model, top-FCNC transitions are forbidden at tree level and remain highly suppressed at loop level by the GIM mechanism; any observable signal would therefore be a clean indication of physics beyond the Standard Model.
- The key rare-top vertices are $tq\gamma$ and tqZ with $q = u, c$, naturally parameterized by anomalous couplings or SMEFT dimension-6 operators.
- At LHμC, the large top sample, harder kinematics, and characteristic μp topology make anomalous single-top and associated production natural benchmarks for converting top statistics into BSM reach.

Top-FCNC at LHμC



Representative process: $\mu^+ q \to \mu^+ t$, with $q = u, c$. Neutral-current exchange probes anomalous $tq\gamma$ and tqZ couplings.

Improved sensitivity relative to the LHeC is expected at LHμC, but quantitative limits remain channel-dependent and require realistic event selections, detector modelling, and systematic-uncertainty estimates.

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Why LHμC is also a distinctive BSM machine

- Muon–proton collisions combine a point-like lepton beam, high centre-of-mass energy, and the possibility of **resonant single production** of new states that are more difficult to isolate at hadron colliders.
- The LHμC is naturally powerful for BSM scenarios tied to second-generation leptons.
- Representative examples include:
  - R-parity-violating squark searches,
  - color-octet muons (leptogluons),
  - excited leptons,
  - contact interactions,
  - and other muon-coupled new-physics scenarios.

Kingman Cheung and Zeren Simon Wang, *Physics potential of a muon-proton collider*, Phys. Rev. D 103 (2021) 116009.

Representative BSM benchmarks: RPV SUSY and color-octet muons

- **RPV SUSY:** The μp collisions can probe resonantly produced squarks with coupling reaches stronger than HL-LHC bounds in the on-shell regime.
- **Color-octet muons:** A 5σ discovery reach of about:
  - $\sim$ 2.3 TeV at the HL-LHC,
  - $\sim$ 3.0 TeV for the 500 GeV LHμC option,
  - $\sim$ 4.2 TeV for the 1 TeV LHμC option.
- These examples are sufficient to demonstrate that the BSM case is not generic hype: it is tied to concrete channels where the μp initial state is a genuine advantage.

Kingman Cheung and Zeren Simon Wang, *Physics potential of a muon-proton collider*, Phys. Rev. D 103 (2021) 116009.

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FCC-hh era outlook and final take-home points

- The long-term perspective goes well beyond the 3.7 TeV baseline.
- The FCC-hh era is explicitly connected to a future high-energy μp option, turning the current LHμC concept into a stepping stone toward even more powerful HET and BSM programmes.
- In particular, a future FCC-μh machine would extend:
  - Higgs statistics and coupling reach,
  - top and rare-top production,
  - high-mass electroweak and $\gamma\gamma$ studies,
  - and resonant BSM sensitivity.

HET@LHμC

(1) LHμC is a serious HET factory;
(2) the **top sector** is one of its clearest near-term strengths;
(3) Higgs, EW, EFT, and $\gamma\gamma$ studies are all strongly motivated;
(4) the concept also provides a natural bridge toward FCC-era μp physics.

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The HELP4CERN collaboration

HELP4CERN collaboration focuses on the physics case for high-energy, high-luminosity lepton-hadron colliders at CERN.

- assess the physics potential of future lepton-hadron collider options,
- develop benchmark studies for Higgs, electroweak, top, QCD, and BSM physics,
- provide a common framework for phenomenology, detector studies, and future projections,
- new contributions and physics studies are very welcome,

Contact: `help-4-cern@cern.ch`
Visit: `https://help4cern.web.cern.ch`

White Paper on High Energy, High Luminosity Lepton-Hadron Colliders at CERN

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